Solutions to the Exercises of Chapter 8

Periodic Orbits and Limit Sets

Chaos: An Introduction to Dynamical Systems, Alligood–Sauer–Yorke

Conventions and results used

Throughout, $\dot{\mathbf{v}} = \mathbf{f}(\mathbf{v})$ is autonomous, $\mathbf{F}(t, \mathbf{v}_0)$ is its flow, and $\mathbf{v}(t) = \mathbf{F}(t, \mathbf{v}_0)$. I use the following facts from the chapter (numbering as in the book).

- **Properties of ω-limit sets.** For a bounded forward orbit, $\omega(\mathbf{v}_0)$ is nonempty, closed, connected, and invariant under the flow (the whole orbit of each of its points, forward *and* backward, lies in it); and $\mathbf{z} \in \omega(\mathbf{u}),\ \mathbf{u} \in \omega(\mathbf{v}_0) \Rightarrow \mathbf{z} \in \omega(\mathbf{v}_0)$. Also $\alpha(\mathbf{v}_0)$ for $\dot{\mathbf{v}} = \mathbf{f}(\mathbf{v})$ is $\omega(\mathbf{v}_0)$ for $\dot{\mathbf{v}} = -\mathbf{f}(\mathbf{v})$ (Exercise 8.13).
- **Theorem 7.16** (continuous dependence): $|\mathbf{F}(t, \mathbf{v}) - \mathbf{F}(t, \mathbf{w})| \le |\mathbf{v} - \mathbf{w}|e^{L|t|}$, L a Lipschitz constant of $\mathbf{f}$.
- **Transversal** at a nonequilibrium point $\mathbf{u}$: a short segment $L \ni \mathbf{u}$, perpendicular to $\mathbf{f}(\mathbf{u})$, containing no equilibria, with $\mathbf{f}$ nowhere tangent to L. All orbits cross L in the same direction.
- **Lemma 8.11 (Diamond Construction).** For small δ, every orbit through a point of the interior of the δ-diamond $D_\delta(\mathbf{u})$ has a segment inside $D_\delta(\mathbf{u})$ that meets L.
- **Lemma 8.12 (Monotonicity).** Successive crossings of a transversal L by one orbit are monotone along L.
- **Lemma 8.13 (Locally a curve).** If $\mathbf{u} \in \omega(\mathbf{v}_0)$ is not an equilibrium, then $\omega(\mathbf{v}_0) \cap D_\delta(\mathbf{u})$ is a single trajectory segment.
- **Corollary 8.14, Lemma 8.15 ("Has one, is one"), Exercise T8.7** (if $\mathbf{v}_0 \in \omega(\mathbf{v}_0)$ then $\omega(\mathbf{v}_0)$ is an equilibrium or a periodic orbit).
- **Theorem 8.8 (Poincaré–Bendixson).** Planar, equilibria isolated, forward orbit bounded: $\omega(\mathbf{v}_0)$ is (1) an equilibrium, (2) a periodic orbit, or (3) a set such that every $\mathbf{u} \in \omega(\mathbf{v}_0)$ has $\alpha(\mathbf{u}), \omega(\mathbf{u})$ equilibria (equilibria plus connecting arcs).

A few solutions refer to others (8.4 uses 8.12(c); 8.12 uses 8.16 and 8.18; 8.11 and 8.12 use 8.18).

Exercise 8.1

Explain why $x(t) = \sin t$ cannot be the solution of a one-dimensional equation $\dot{x} = f(x)$.

Solution. Suppose $\dot{x} = f(x)$ had the solution $x(t) = \sin t$. Then $f(\sin t) = \cos t$ for all t. Since $\sin(\pi/6) = \sin(5\pi/6) = \frac{1}{2}$, we would need

$$f(\tfrac{1}{2}) = \cos\tfrac{\pi}{6} = \tfrac{\sqrt{3}}{2} \qquad \text{and} \qquad f(\tfrac{1}{2}) = \cos\tfrac{5\pi}{6} = -\tfrac{\sqrt{3}}{2},$$

which is impossible for a single-valued function f. (Equivalently, by Theorem 8.3 a nonconstant solution of a scalar autonomous equation is monotonic, but $\sin t$ has a maximum at $t = \pi/2$; the maximum value 1 would be an equilibrium, and by uniqueness the solution would be constant.)

Exercise 8.2

Find the ω-limit sets of $\dot{r} = r(r-1)(3-r)$, $\dot{\theta} = 1$ for $(r,\theta) = (0,0), (\frac{1}{2},0), (1,0), (2,0)$.

Solution. The r-equation is decoupled. Its equilibria are $r = 0, 1, 3$, and

$$\dot{r} < 0 \text{ on } (0,1), \qquad \dot{r} > 0 \text{ on } (1,3), \qquad \dot{r} < 0 \text{ on } (3,\infty).$$

So $r = 0$ and $r = 3$ are attracting and $r = 1$ is repelling; θ increases at unit rate, so the orbits turn counterclockwise.

- $(0,0)$ is the origin, an equilibrium: $\omega = \{(0,0)\}$.
- $(\frac{1}{2},0)$: $r(t)$ decreases monotonically and tends to 0 (it cannot reach 0 in finite time by uniqueness). The orbit spirals into the origin: $\omega = \{(0,0)\}$.
- $(1,0)$: $r \equiv 1$, $\theta = t$, a periodic orbit. $\omega =$ the circle $r = 1$.
- $(2,0)$: $r(t) \uparrow 3$ and $\theta(t) = t$. For any angle φ choose $t_n \to \infty$ with $\theta(t_n) \equiv \varphi \pmod{2\pi}$; then $(r,\theta)(t_n) \to (3,\varphi)$. So $\omega =$ the circle $r = 3$.

Exercise 8.3

A is a real 2×2 matrix. (a) Describe the ω-limit sets of $\dot{\mathbf{v}} = A\mathbf{v}$. (b) When are the equilibria isolated? (c) Repeat for 3×3.

Solution. (a) Here $\mathbf{v}(t) = e^{At}\mathbf{v}_0$ and $\omega(\mathbf{v}_0)$ is read off from the canonical form of A (eigenvalues $\lambda_{1,2}$):

Eigenvalues	$\boldsymbol{\omega(\mathbf{v}_0)}$
$\operatorname{Re}\lambda_{1,2} < 0$ (sink; includes repeated eigenvalues/Jordan blocks)	$\{\mathbf{0}\}$ for every $\mathbf{v}_0$
$\lambda_1 < 0 < \lambda_2$ (saddle)	$\{\mathbf{0}\}$ if $\mathbf{v}_0$ lies on the stable eigenline; $\emptyset$ otherwise (orbit unbounded)
$\operatorname{Re}\lambda_{1,2} > 0$ (source)	$\{\mathbf{0}\}$ if $\mathbf{v}_0 = \mathbf{0}$; $\emptyset$ otherwise
$\lambda = \pm i\beta$, $\beta \neq 0$ (center)	$\{\mathbf{0}\}$ if $\mathbf{v}_0 = \mathbf{0}$; otherwise the periodic orbit (ellipse) through $\mathbf{v}_0$
$\lambda_1 = 0,\ \lambda_2 = \mu < 0$	one equilibrium: if $\mathbf{v}_0 = c_1\mathbf{u}_1 + c_2\mathbf{u}_2$ ($\mathbf{u}_1 \in \ker A$, $\mathbf{u}_2$ an eigenvector for μ) then $\omega(\mathbf{v}_0) = \{c_1\mathbf{u}_1\}$
$\lambda_1 = 0,\ \lambda_2 = \mu > 0$	$\{\mathbf{v}_0\}$ if $\mathbf{v}_0 \in \ker A$; $\emptyset$ otherwise
$\lambda_1 = \lambda_2 = 0$	if $A = 0$: $\{\mathbf{v}_0\}$; if $A \neq 0$ (nilpotent), $\mathbf{v}(t) = \mathbf{v}_0 + tA\mathbf{v}_0$: $\{\mathbf{v}_0\}$ if $\mathbf{v}_0 \in \ker A$, else $\emptyset$

In every case $\omega(\mathbf{v}_0)$ is empty, a single equilibrium, or a periodic orbit (the latter only for a center, where it is the orbit itself).

(b) The equilibria are the solutions of $A\mathbf{v} = \mathbf{0}$, i.e. $\ker A$. This is $\{\mathbf{0}\}$ (so the equilibrium is isolated) iff $\det A \neq 0$, i.e. iff 0 is not an eigenvalue. If $\det A = 0$, $\ker A$ is a line or the whole plane, and equilibria are not isolated.

(c) For 3×3 the trichotomy is the same: $\omega(\mathbf{v}_0)$ is empty, a single equilibrium, or a periodic orbit. Equilibria are isolated iff $\det A \neq 0$ (no zero eigenvalue); otherwise $\ker A$ is a line, plane or $\mathbb{R}^3$. A real 3×3 matrix has either three real eigenvalues or one real λ and a pair $\alpha \pm i\beta$, so there is at most *one* rotation frequency and nothing like a torus can occur. The new feature compared with 2×2: a nonperiodic orbit can have a periodic ω-limit set. Example: $A = \mathrm{diag}(-1, J)$ with $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, i.e. $\dot{z} = -z$, $(\dot{x}, \dot{y}) = (-y, x)$. Then $\omega(x_0, y_0, z_0)$ is the circle $x^2 + y^2 = x_0^2 + y_0^2$, $z = 0$ (if $(x_0, y_0) \neq (0, 0)$). In general, when A has eigenvalues $\pm i\beta$ and a real λ: for $\lambda < 0$ every orbit has the periodic orbit in the rotation plane as its ω-limit; for $\lambda = 0$ every orbit is itself periodic (or an equilibrium); for $\lambda > 0$ only orbits in the rotation plane are bounded. All other eigenvalue patterns give empty ω-limit sets or a single equilibrium as in the 2×2 table.

Exercise 8.4

A planar equation has a single equilibrium $\mathbf{v}_0$, which is attracting. Describe the possible limit sets of a trajectory $\mathbf{v}(t)$ that is bounded for $t \geq 0$ but does not converge to $\mathbf{v}_0$.

Solution. "Attracting" means there is a neighborhood W of $\mathbf{v}_0$ all of whose points converge to $\mathbf{v}_0$. The only equilibrium is $\mathbf{v}_0$, so it is isolated and the Poincaré–Bendixson Theorem applies to $\omega(\mathbf{v})$.

Claim: $\mathbf{v}_0 \notin \omega(\mathbf{v})$. If $\mathbf{v}_0 \in \omega(\mathbf{v})$, then $\mathbf{v}(t_n) \to \mathbf{v}_0$ for some $t_n \to \infty$, so $\mathbf{v}(t_n) \in W$ for large n and then $\mathbf{v}(t) = \mathbf{F}(t - t_n, \mathbf{v}(t_n)) \to \mathbf{v}_0$, contradicting the hypothesis.

Now go through the three cases of Theorem 8.8.

(1) $\omega(\mathbf{v}) = \{\mathbf{v}_0\}$ is excluded by the claim.

(2) $\omega(\mathbf{v})$ is a periodic orbit.

(3) Every $\mathbf{u} \in \omega(\mathbf{v})$ has $\omega(\mathbf{u}) = \{\mathbf{v}_0\}$. But $\omega(\mathbf{u}) \subset \omega(\mathbf{v})$ (closure and invariance, or transitivity), so $\mathbf{v}_0 \in \omega(\mathbf{v})$, again excluded.

Hence the **only possibility is that $\omega(\mathbf{v})$ is a periodic orbit**, a limit cycle Γ. By Exercise 8.12(c), Γ must enclose an equilibrium, which can only be $\mathbf{v}_0$. So Γ is a closed orbit surrounding $\mathbf{v}_0$; there are no connecting arcs or homoclinic loops (they would have to pass through $\mathbf{v}_0$).

Exercise 8.5

Find the α- and ω-limit sets for the flows in Figure 8.4.

Solution. In each panel an equilibrium, or the set listed, is its own limit set; below I list the nontrivial orbits. An orbit which comes in from infinity in backward time has $\alpha = \emptyset$.

(a) The origin O is a spiral sink; orbits spiral in and (backward in time) spiral out to infinity. For every orbit $\neq O$: $\omega = \{O\}$, $\alpha = \emptyset$. For O: $\alpha = \omega = \{O\}$.

(b) The origin is a spiral source, the circle C is an attracting periodic orbit.

- Orbit starting near O and spiraling out to C: $\alpha = \{O\}$, $\omega = C$.
- Orbit coming from outside and spiraling in to C: $\alpha = \emptyset$, $\omega = C$.
- C itself: $\alpha = \omega = C$. The origin: $\alpha = \omega = \{O\}$.

(c) P is a saddle with two connecting arcs (homoclinic loops) Γ_1, Γ_2 forming the figure eight $\Gamma = \{P\} \cup \Gamma_1 \cup \Gamma_2$; inside each lobe there is a spiral sink S_1, S_2 (reading the picture so that nested curves wind into the dots).

- Outermost orbit (outside the eight): $\omega = \Gamma$, $\alpha = \emptyset$.
- Points on the arcs Γ_1, Γ_2: $\alpha = \omega = \{P\}$.
- An orbit inside lobe i (other than S_i): $\omega = \{S_i\}$, $\alpha = \{P\} \cup \Gamma_i$ (the loop around that lobe).
- P, S_1, S_2: each is its own α- and ω-limit set.

(If the arrows inside the lobes were reversed, α and ω would swap there.)

Exercise 8.6 and 8.7

Which of the letters A B C D P Q Y can be ω-limit sets of planar flows, (8.6) assuming equilibria are isolated, (8.7) without that assumption?

Solution. Answer. (8.6) only **D**. (8.7) **A, C, D, P, Y** (not B, not Q).

I take the letters in standard block type: **A**: two slanted strokes meeting at an apex, joined by a crossbar. **B**: a stem with two bowls that share one middle bar, i.e. a "theta" graph. **C**: an open arc. **D**: stem plus bowl, a simple closed curve. **P**: a bowl on a stem that sticks out below it. **Q**: an oval with a short tail crossing it. **Y**: three strokes meeting at a point.

Obstruction 1 (one complementary component). *Let $\Lambda = \omega(\mathbf{v}_0)$ be the limit set of a bounded planar orbit. If Λ is neither an equilibrium nor a periodic orbit, then $\Lambda \subset \partial U$ for a single connected component U of $\mathbb{R}^2 \setminus \Lambda$.*

Proof. If $\mathbf{v}_0 \in \Lambda$ then Λ is an equilibrium or periodic orbit (T8.7). Otherwise the orbit of $\mathbf{v}_0$ never meets Λ (if $\mathbf{v}(t_1) \in \Lambda$, invariance would put $\mathbf{v}_0 = \mathbf{F}(-t_1, \mathbf{v}(t_1))$ in Λ). The forward orbit is connected and misses Λ, so it lies in one component U of $\mathbb{R}^2 \setminus \Lambda$. Each point of Λ is a limit of orbit points, so $\Lambda \subset \overline{U}$, and $\Lambda \cap U = \emptyset$ gives $\Lambda \subset \partial U$. □

(A periodic orbit satisfies this as well: it is the boundary of its inside and of its outside.)

Obstruction 2 (no free ends, isolated equilibria). *If equilibria are isolated and Λ is neither an equilibrium nor a periodic orbit, then Λ has no "free end".* By Theorem 8.8 (case 3), Λ consists of equilibria and connecting arcs. Suppose Λ ended at a point p (so near p, Λ is a single arc with tip p). Since Λ is invariant, p must be an equilibrium. An orbit that approximates Λ must run along the arc to the tip and then come back along the arc (there is nothing else near p to follow). Take a transversal L at a nonequilibrium point z of that arc: the orbit crosses L once heading toward the tip and once heading away from it, i.e. in opposite directions. But all orbits cross a transversal in the same direction. Contradiction. (This is the argument of Example 8.16(b).)

Letter by letter.

	Complementary components and boundaries	**8.6**	**8.7**
A	outside: boundary is all of A (the crossbar is seen from below); triangular hole	no (free leg ends)	yes
B	outside: boundary omits the middle bar; each bowl's boundary is only its own loop. No component has boundary all of B	no	no
C	one component, boundary is all of C	no (free ends)	yes
D	inside and outside, boundary is all of D	yes (periodic orbit)	yes
P	outside: boundary is all of P (tail is outside the bowl)	no (free tail end)	yes
Q	the part of the tail inside the oval borders only the inside, the part outside borders only the outside	no	no
Y	one component, boundary is all of Y	no (free ends)	yes

8.6. B and Q violate Obstruction 1; A, C, P, Q, Y have free ends and violate Obstruction 2. D is a simple closed curve; it is the ω-limit set of a spiraling orbit around a periodic orbit (Example 8.5). *(If B were drawn as two bowls touching in a single point, it would be a figure eight, which* is *possible, Figure 8.4(c); with a shared middle bar it is not.)*

8.7. Without isolation, arcs may consist entirely of equilibria, so free ends are no longer an obstruction; Obstruction 1 remains, so B and Q are impossible. Conversely, each of A, C, D, P, Y can occur. *Construction (sketch).* Let K be the letter and U the unbounded component of $\mathbb{R}^2 \setminus K$; every point of K lies in ∂U. Choose a smooth simple curve $\gamma : [0, \infty) \to U$ that winds around K in nested laps, lap n following the outer outline of K at distance less than $1/n$ (joining lap n to lap $n+1$ by a short segment). Then γ is closed in $\mathbb{R}^2 \setminus K$ and its accumulation set is exactly K. Take a nowhere-vanishing smooth field V on $\mathbb{R}^2 \setminus K$ tangent to γ, and a smooth $g \geq 0$ with $g^{-1}(0) = K$, so flat at K that gV extends smoothly by 0 across K. The field $\mathbf{f} = gV$ makes every point of K an equilibrium, γ is an orbit (reparametrized), and since $\mathbf{f}$ is bounded and γ has infinite length, γ takes infinite time to traverse. Hence $\omega(\gamma(0)) = K$.

Exercise 8.8

Find the ω-limit sets for the phase planes of $\ddot{x} + P'(x) = 0$ for the potentials of Figure 7.26.

Solution. Write $\dot{x} = y$, $\dot{y} = -P'(x)$. The energy $E(x, y) = \frac{1}{2}y^2 + P(x)$ is conserved, so orbits lie on level curves of E. As drawn, $P(x) \to \infty$ as $|x| \to \infty$, so every level set is compact and all orbits are bounded. Equilibria are $(x^*, 0)$ with $P'(x^*) = 0$.

(a) P has local minima at x_1, x_2 and a local maximum at some x_3 with $x_1 < x_3 < x_2$. Then $(x_1, 0), (x_2, 0)$ are centers and $(x_3, 0)$ is a saddle. Put $c = P(x_3)$.

- $E < c$: the level set has two components, one around each center; each point lies on a periodic

orbit and ω = that periodic orbit (the level curve). The centers: $\omega = \{(x_i, 0)\}$.

- $E > c$: one closed curve around both centers; ω = that periodic orbit.
- $E = c$: the level set is a figure eight with the saddle at the crossing; the two lobes minus the saddle are two homoclinic orbits. For a point on either one, $\alpha = \omega = \{(x_3, 0)\}$; for the saddle, $\omega = \{(x_3, 0)\}$.

(b) P has a minimum at x_1 and an inflection point at $x_2 > x_1$ with $P'(x_2) = 0$ (and P increasing through it). Then $(x_1, 0)$ is a center and $(x_2, 0)$ is a degenerate (cusp-type) equilibrium. Put $c = P(x_2)$.

- $E < c$: periodic orbits around $(x_1, 0)$; ω = the orbit itself.
- $E = c$: the level set is a single loop returning to $(x_2, 0)$ with a cusp; minus that point it is one homoclinic orbit, so $\alpha = \omega = \{(x_2, 0)\}$ for points on it.
- $E > c$: periodic orbits enclosing the loop; ω = the orbit itself.

In both cases the only ω-limit sets are equilibria, periodic orbits and (via homoclinic orbits) the saddle/cusp point; no orbit has the figure eight or the cusp loop as its ω-limit set, since energy is conserved.

Exercise 8.9

Let $\ddot{x} + b\dot{x} + g(x) = 0$, $b > 0$. (a) Show there are no non-equilibrium periodic solutions. (b) Find the ω-limit sets for $\ddot{x} + b\dot{x} + P'(x) = 0$, P as in Figure 7.26.

Solution. (a) Let $y = \dot{x}$ and $E(x, y) = \frac{1}{2}y^2 + \int^x g$. Then

$$\dot{E} = y\dot{y} + g(x)\dot{x} = y(-by - g(x)) + g(x)y = -by^2 \le 0,$$

with equality only when $y = 0$. Suppose $(x(t), y(t))$ is periodic with period T. Then E returns to its value, so $0 = E(T) - E(0) = -b\int_0^T y^2\,dt$, so $y \equiv 0$, hence x is constant: an equilibrium. So there are no nonconstant periodic solutions.

(b) With $g = P'$ and $P \to \infty$ at $\pm\infty$ (as drawn), the sets $\{E \le c\}$ are compact and forward invariant, so every forward orbit is bounded. $E(\mathbf{v}(t))$ is nonincreasing and bounded below, so tends to a constant c_0, and $E \equiv c_0$ on $\omega(\mathbf{v}_0)$. As $\omega(\mathbf{v}_0)$ is invariant, E is constant along the orbits in it, so $\dot{E} = -by^2 = 0$ there, i.e. $y \equiv 0$ along each such orbit; then x is constant and $P'(x) = 0$. Hence $\omega(\mathbf{v}_0)$ consists of equilibria. It is connected and the equilibria are isolated, so $\omega(\mathbf{v}_0)$ **is a single equilibrium for every** $\mathbf{v}_0$ (no periodic orbits, no connecting arcs). The equilibria have Jacobian $\left(\begin{smallmatrix} 0 & 1 \\ -P'' & -b \end{smallmatrix}\right)$, with characteristic polynomial $\lambda^2 + b\lambda + P''(x^*)$.

(a) $(x_1, 0)$ and $(x_2, 0)$ (minima of P, $P'' > 0$) are sinks. $(x_3, 0)$ is a saddle ($P''(x_3) < 0$). Thus $\omega(\mathbf{v}_0) = \{(x_3, 0)\}$ exactly when $\mathbf{v}_0$ lies on the stable manifold of the saddle (a curve with two branches, together with the saddle); for every other $\mathbf{v}_0$, $\omega(\mathbf{v}_0)$ is $\{(x_1, 0)\}$ or $\{(x_2, 0)\}$. The two basins are open sets separated by the stable manifold of the saddle.

(b) $(x_1, 0)$ is a sink. At $(x_2, 0)$, $P'(x_2) = P''(x_2) = 0$, so the eigenvalues are 0 and $-b$: a saddle–node. On the center manifold $\dot{x} \approx -P'(x)/b \le 0$, so the equilibrium attracts from the side $x > x_2$ and repels on the side $x < x_2$. Hence $\omega(\mathbf{v}_0) = \{(x_2, 0)\}$ for $\mathbf{v}_0 = (x_2, 0)$, for points on the strong stable curve, and for points in the open "node" sector on the right (these all have $E(\mathbf{v}_0) > P(x_2)$, since E decreases to $P(x_2)$); for every other $\mathbf{v}_0$, $\omega(\mathbf{v}_0) = \{(x_1, 0)\}$.

Exercise 8.10

State a version of the Poincaré–Bendixson Theorem without the isolated-equilibria hypothesis, explain how to adapt the proof, and decide whether a limit set containing a periodic orbit must equal it.

Solution. Theorem. Let $\mathbf{f}$ be a smooth planar vector field and suppose the forward orbit of $\mathbf{v}_0$ is bounded. Then one of the following holds.

(1) $\omega(\mathbf{v}_0)$ consists entirely of equilibria (it is then a compact connected set of equilibria: a point or a continuum).

(2) $\omega(\mathbf{v}_0)$ is a periodic orbit.

(3) $\omega(\mathbf{v}_0)$ contains non-equilibrium points, and for each non-equilibrium $\mathbf{u} \in \omega(\mathbf{v}_0)$ the sets $\alpha(\mathbf{u})$ and $\omega(\mathbf{u})$ each consist of equilibria (each a connected set of equilibria contained in $\omega(\mathbf{v}_0)$).

Adapting the proof. Isolation of equilibria was used only to pass from "consists of equilibria" to "is a single equilibrium". The lemmas 8.11–8.15, T8.7 and Corollary 8.14 involve only nonequilibrium points (transversals and diamonds are built around such points, and a small neighborhood of a nonequilibrium contains no equilibria since $\mathbf{f}$ is continuous), so they go through unchanged. The proof becomes:

- If $\omega(\mathbf{v}_0)$ contains only equilibria, case (1) holds (connectedness gives a connected set of equilibria, not necessarily a point).
- Otherwise take a nonequilibrium $\mathbf{u} \in \omega(\mathbf{v}_0)$. If $\omega(\mathbf{u})$ contains a nonequilibrium $\mathbf{z}$, then Corollary 8.14 shows $\mathbf{u}$ lies on a periodic orbit, and by Lemma 8.15, $\omega(\mathbf{v}_0)$ is a periodic orbit: case (2). The same holds for $\alpha(\mathbf{u})$: since $\alpha(\mathbf{u}) \subset \omega(\mathbf{v}_0)$ (invariance and closure), a nonequilibrium $\mathbf{z} \in \alpha(\mathbf{u})$ has the orbit of $\mathbf{u}$ passing arbitrarily close to $\mathbf{z}$ inside $\omega(\mathbf{v}_0)$, so by Lemma 8.13 the orbit of $\mathbf{u}$ is the orbit of $\mathbf{z}$, so $\mathbf{z} \in \alpha(\mathbf{z})$, and the time-reversed version of T8.7 makes $\mathbf{u}$ periodic; then again $\omega(\mathbf{v}_0)$ is that periodic orbit.
- If neither $\omega(\mathbf{u})$ nor $\alpha(\mathbf{u})$ contains a nonequilibrium (for every such $\mathbf{u}$), they are connected sets of equilibria: case (3).

Must the periodic orbit be the whole limit set? Yes. The proof of Lemma 8.15 ("Has one, is one") uses only connectedness of $\omega(\mathbf{v}_0)$, compactness of the periodic orbit G, and Lemma 8.13 at a point of G (a nonequilibrium point). It never uses isolation of equilibria. So if $\omega(\mathbf{v}_0)$ contains a periodic orbit G, then $\omega(\mathbf{v}_0) = G$; in particular a limit set cannot be a periodic orbit together with extra equilibria or arcs.

Exercise 8.11

Use Continuous Dependence (Theorem 7.16) to give an alternate proof of Lemma 8.15.

Solution. Let $\mathbf{v}(t) = \mathbf{F}(t, \mathbf{v}_0)$ have bounded forward orbit and suppose a periodic orbit G, of period T, lies in $\omega(\mathbf{v}_0)$. We show $\omega(\mathbf{v}_0) = G$. If the orbit of $\mathbf{v}_0$ is itself periodic then $\omega(\mathbf{v}_0)$ is that orbit and contains G, so equals G; so assume it is not periodic.

Fix $\mathbf{u} \in G$ and a transversal L at $\mathbf{u}$, short enough that $L \cap G = \{\mathbf{u}\}$. Let $a = |\mathbf{f}(\mathbf{u})|$.

Step 1: the orbit crosses L at points $c_k \to \mathbf{u}$. Since $\mathbf{u} \in \omega(\mathbf{v}_0)$, there are $t_n \to \infty$ with $\mathbf{v}(t_n) \to \mathbf{u}$. By the Diamond Construction (with $\delta_n \to 0$) the orbit through $\mathbf{v}(t_n)$ crosses L at a time s_n with

$|s_n - t_n| \le C\delta_n$ and $|\mathbf{v}(s_n) - \mathbf{u}| \le \delta_n$. So the orbit crosses L at times $s_n \to \infty$ at points converging to $\mathbf{u}$. By Exercise 8.18, in any bounded time interval there are only finitely many crossings, so we can list all crossing times $q_1 < q_2 < \cdots \to \infty$, with points $c_k = \mathbf{v}(q_k) \in L$. By Lemma 8.12 (and since the orbit is nonperiodic, the c_k are distinct) (c_k) is a monotone sequence along L. A monotone sequence with a subsequence converging to $\mathbf{u}$ converges to $\mathbf{u}$: $c_k \to \mathbf{u}$.

Step 2: the return time. Let $z = c_k$ with k large. Since $\mathbf{F}(T, \mathbf{u}) = \mathbf{u}$, Theorem 7.16 gives $|\mathbf{F}(T, z) - \mathbf{u}| \le |z - \mathbf{u}|e^{LT}$, which is small; so $\mathbf{F}(T, z) \in D_\delta(\mathbf{u})$ and the Diamond Construction says the orbit of z crosses L at some time $T + \sigma$ with $|\sigma| \le C\delta$. For $t \in [\rho, T - \rho]$, the point $\mathbf{F}(t, \mathbf{u}) \in G$ stays a positive distance $m(\rho)$ from L (because $G \cap L = \{\mathbf{u}\}$); by Theorem 7.16, if $|z - \mathbf{u}|e^{LT} < m(\rho)$ the orbit of z does not meet L on $[\rho, T - \rho]$. Near the two ends of the period it is inside the diamond, where it meets L only once. Hence for large k the *next* crossing after q_k is at $q_{k+1} = q_k + T + \sigma_k$, $|\sigma_k| \le C\delta$.

Step 3: the orbit stays close to G. For $t \in [q_k, q_{k+1}]$, Theorem 7.16 applied to the orbits through c_k and $\mathbf{u}$ (and $\mathbf{F}(t, \mathbf{u}) \in G$ for all t) gives

$$\operatorname{dist}(\mathbf{v}(t), G) \le |c_k - \mathbf{u}|\, e^{L(T+1)} \xrightarrow[k\to\infty]{} 0.$$

So $\operatorname{dist}(\mathbf{v}(t), G) \to 0$ as $t \to \infty$, i.e. every point of $\omega(\mathbf{v}_0)$ lies in G. With $G \subset \omega(\mathbf{v}_0)$, we get $\omega(\mathbf{v}_0) = G$. □

(Planarity enters only through Lemma 8.12, which is what makes *all* later crossings converge to $\mathbf{u}$; in $\mathbb{R}^3$ the statement is false.)

Exercise 8.12

Planar vector field. (a) If $\alpha(\mathbf{p})$ and $\omega(\mathbf{p})$ are the same periodic orbit, prove $\mathbf{p}$ lies on it. (b) Width of orbits and periodic orbits. (c) Any planar field with a periodic orbit has an equilibrium.

Solution. (a) Let $G = \alpha(\mathbf{p}) = \omega(\mathbf{p})$ be periodic and write $\mathbf{v}(t) = \mathbf{F}(t, \mathbf{p})$. If the orbit of $\mathbf{p}$ is periodic, it equals its own ω-limit set, so equals G and $\mathbf{p} \in G$. So suppose, for contradiction, that the orbit of $\mathbf{p}$ is not periodic.

Pick $\mathbf{u} \in G$ and a transversal L at $\mathbf{u}$ with $L \cap G = \{\mathbf{u}\}$. Since $\mathbf{u} \in \omega(\mathbf{p})$ and $\mathbf{u} \in \alpha(\mathbf{p})$, the Diamond Construction gives crossings of L at times going to $+\infty$ and to $-\infty$. By Exercise 8.18 the crossing times form a doubly infinite increasing sequence $\cdots < \tau_{-1} < \tau_0 < \tau_1 < \cdots$, with crossing points $c_k = \mathbf{v}(\tau_k) \in L$, all distinct (the orbit is not periodic). By Lemma 8.12 the sequence $(c_k)_{k\in\mathbb{Z}}$ is strictly monotone along L. Limit points of c_k as $k \to +\infty$ lie in $\omega(\mathbf{p}) \cap L = G \cap L = \{\mathbf{u}\}$, so $c_k \to \mathbf{u}$; likewise $c_k \to \mathbf{u}$ as $k \to -\infty$. But a strictly monotone bi-infinite sequence has different limits at $+\infty$ and $-\infty$. Contradiction. Hence $\mathbf{p} \in G$.

(b) Let G be a periodic orbit of width w (its diameter; $0 < w < \infty$). By the Jordan Curve Theorem G bounds a closed region D whose interior is nonempty, and D is invariant (orbits cannot cross G). We have $\operatorname{diam} D = w$: if $\mathbf{e} \in \operatorname{int} D$ and $\mathbf{b} \in D$, then moving $\mathbf{e}$ slightly away from $\mathbf{b}$ stays in D and increases the distance, so a pair realizing $\operatorname{diam} D$ must lie on $\partial D = G$.

Take $\mathbf{p} \in \operatorname{int} D$. Both its forward and backward orbits lie in $\operatorname{int} D$, so their widths are at most w. Suppose the forward width equals w. The closure K of the forward orbit is $K = \{\mathbf{F}(t, \mathbf{p}) : t \ge 0\} \cup \omega(\mathbf{p}) \subset D$, is compact, and has diameter $w = \operatorname{diam} D$; so there are $\mathbf{a}, \mathbf{b} \in K$ with $|\mathbf{a} - \mathbf{b}| = w$, and by the above $\mathbf{a}, \mathbf{b} \in G$. As $G \cap \{\mathbf{F}(t, \mathbf{p})\} = \emptyset$, $\mathbf{a}, \mathbf{b} \in \omega(\mathbf{p})$. By invariance $G \subset \omega(\mathbf{p})$, and by Lemma 8.15, $\omega(\mathbf{p}) = G$. Likewise, if the backward width equals w then $\alpha(\mathbf{p}) = G$ (apply the same

argument to $-\mathbf{f}$). If both held, part (a) would give $\mathbf{p} \in G$, contradicting $\mathbf{p} \in \operatorname{int} D$. So the forward or the backward orbit of $\mathbf{p}$ has width $< w$.

(c) Let G be a periodic orbit and D the closed region it bounds; D is homeomorphic to a closed disc (Schoenflies) and is invariant, so each time-t map $\mathbf{F}(t,\cdot) : D \to D$ is continuous. By Brouwer's fixed point theorem, for each n there is $\mathbf{p}_n \in D$ with $\mathbf{F}(1/n, \mathbf{p}_n) = \mathbf{p}_n$. Either $\mathbf{p}_n$ is an equilibrium, or it lies on a periodic orbit whose period T_n divides $1/n$, so $T_n \le 1/n$. But $\mathbf{f}$ is Lipschitz on the compact set D with some constant L, and by Exercise 8.16 every periodic orbit in D has period at least $2\pi/L$. For large n this is impossible, so $\mathbf{p}_n$ is an equilibrium. Thus D contains an equilibrium. □

(One can instead derive (c) from (b) by iterating it to get nested periodic orbits of shrinking width.)

Exercise 8.13

(a) Show that the α-limit set of $\dot{\mathbf{v}} = \mathbf{f}(\mathbf{v})$ is the ω-limit set of $\dot{\mathbf{v}} = -\mathbf{f}(\mathbf{v})$. (b) State and prove an analogue of Poincaré–Bendixson for α-limit sets.

Solution. (a) If $\mathbf{v}(t) = \mathbf{F}(t, \mathbf{v}_0)$ solves $\dot{\mathbf{v}} = \mathbf{f}(\mathbf{v})$, then $\mathbf{w}(t) = \mathbf{v}(-t)$ solves $\dot{\mathbf{w}} = -\mathbf{f}(\mathbf{w})$ with $\mathbf{w}(0) = \mathbf{v}_0$, because $\dot{\mathbf{w}}(t) = -\dot{\mathbf{v}}(-t) = -\mathbf{f}(\mathbf{v}(-t)) = -\mathbf{f}(\mathbf{w}(t))$. Now $\mathbf{z} \in \omega_{-\mathbf{f}}(\mathbf{v}_0)$ iff $\mathbf{w}(t_n) = \mathbf{v}(-t_n) \to \mathbf{z}$ for some $t_n \to +\infty$ iff $\mathbf{v}(s_n) \to \mathbf{z}$ for some $s_n = -t_n \to -\infty$ iff $\mathbf{z} \in \alpha_{\mathbf{f}}(\mathbf{v}_0)$.

(b) Theorem. Let $\mathbf{f}$ be a smooth planar field with isolated equilibria, and suppose the backward orbit $\{\mathbf{F}(t, \mathbf{v}_0) : t \le 0\}$ is bounded. Then either (1) $\alpha(\mathbf{v}_0)$ is an equilibrium; (2) $\alpha(\mathbf{v}_0)$ is a periodic orbit; or (3) for each $\mathbf{u} \in \alpha(\mathbf{v}_0)$ the limit sets $\alpha(\mathbf{u}), \omega(\mathbf{u})$ are equilibria.

Proof. Apply Theorem 8.8 to $-\mathbf{f}$. It is smooth and has the same (isolated) equilibria; the forward orbit of $\mathbf{v}_0$ under $-\mathbf{f}$ is the backward orbit under $\mathbf{f}$, hence bounded. By (a), $\omega_{-\mathbf{f}}(\mathbf{v}_0) = \alpha_{\mathbf{f}}(\mathbf{v}_0)$. Periodic orbits of $-\mathbf{f}$ and $\mathbf{f}$ are the same sets, and for every $\mathbf{u}$, $\alpha_{-\mathbf{f}}(\mathbf{u}) = \omega_{\mathbf{f}}(\mathbf{u})$ and $\omega_{-\mathbf{f}}(\mathbf{u}) = \alpha_{\mathbf{f}}(\mathbf{u})$. Translating the three conclusions of Theorem 8.8 gives (1)–(3). □

Exercise 8.14

If $|\mathbf{v}(t)| \to \infty$ as $t \to \infty$, show $\omega(\mathbf{v}_0) = \emptyset$.

Solution. If $\mathbf{z} \in \omega(\mathbf{v}_0)$, there are $t_n \to \infty$ with $\mathbf{v}(t_n) \to \mathbf{z}$. Then $|\mathbf{v}(t_n)| \to |\mathbf{z}| < \infty$, which contradicts $|\mathbf{v}(t)| \to \infty$. Hence no such $\mathbf{z}$ exists. □

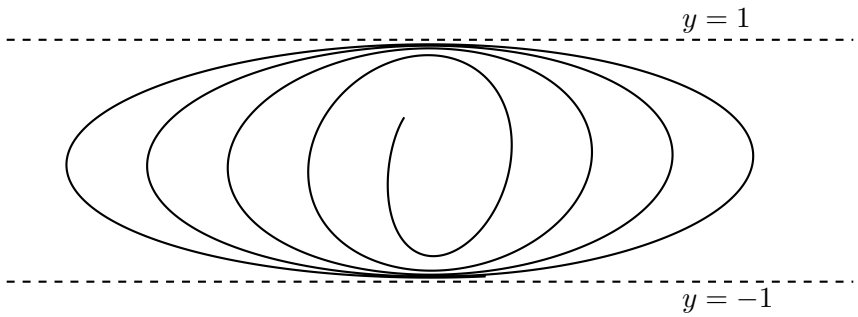
Exercise 8.15

Sketch a planar vector field for which the ω-limit set of a single trajectory is two (unbounded) parallel lines.

Solution. Take the lines $\ell_\pm = \{y = \pm 1\}$. The orbit should sweep out along the top line to the right, return along the bottom line, and repeat, with each lap longer and closer to the lines than the last (like a spiral on an ever-longer, ever-thinner ellipse). An explicit curve of this kind is

$$\gamma(s) = \Big(s\cos s,\ \big(1 - \tfrac{1}{s}\big)\sin s \Big), \qquad s \ge 2.$$

It runs on ellipses with semi-axes s and $1 - 1/s$, both increasing, so the laps are nested and γ is a simple curve. For fixed x_0, points of γ with $s\cos s \to x_0$ have $\cos s \to 0$, hence $y \to \pm 1$; and every $(x_0, \pm 1)$ is approached. Points with $|x| \to \infty$ are not limit points. So the set of limit points of γ is exactly $\ell_+ \cup \ell_-$.



Making it a vector field. The curve γ is a closed subset of the open set $U = \mathbb{R}^2 \setminus (\ell_+ \cup \ell_-)$ and is embedded, so there is a smooth nonvanishing field V on U tangent to γ (take the unit tangent along γ and extend it smoothly to a neighborhood). Choose $g \geq 0$ smooth with $g^{-1}(0) = \ell_+ \cup \ell_-$ and so flat there that $\mathbf{f} = gV$ extends smoothly by 0 to $\mathbb{R}^2$, and bounded. Then the lines consist of equilibria, γ is an orbit, and because $\mathbf{f}$ is bounded and γ has infinite length, the orbit exists for all $t \geq 0$ and $\omega(\gamma(2)) = \ell_+ \cup \ell_-$. Of course the orbit is unbounded, so Poincaré–Bendixson (which requires a bounded orbit) does not apply.

Exercise 8.16

Let L be a Lipschitz constant for $\mathbf{f}$ in $\dot{\mathbf{v}} = \mathbf{f}(\mathbf{v})$ (planar), and $\mathbf{x}(t)$ a periodic orbit with period T. Show that $T \geq 2\pi/L$.

Solution. Assume $\mathbf{f} \in C^1$, so $\|D\mathbf{f}\| \leq L$ (for merely Lipschitz $\mathbf{f}$ the same estimate on the turning rate holds with the upper Dini derivative). The orbit is not an equilibrium, so $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t)) \neq 0$. Since $\ddot{\mathbf{x}} = D\mathbf{f}(\mathbf{x})\dot{\mathbf{x}}$,

$$|\ddot{\mathbf{x}}| \leq L|\dot{\mathbf{x}}|.$$

Write $\dot{\mathbf{x}} = |\dot{\mathbf{x}}|(\cos\theta, \sin\theta)$ with $\theta(t)$ a continuous (indeed C^1) choice of angle. Then

$$\dot{\theta} = \frac{\dot{x}_1\ddot{x}_2 - \dot{x}_2\ddot{x}_1}{|\dot{x}|^2}, \qquad \text{so} \qquad |\dot{\theta}| \leq \frac{|\ddot{\mathbf{x}}|}{|\dot{\mathbf{x}}|} \leq L.$$

Let T be the minimal period (a larger period only makes T larger). By uniqueness, $\mathbf{x}([0,T])$ is a simple closed C^1 curve, so by Hopf's Umlaufsatz (the tangent turning number of a simple closed curve is ± 1), $|\theta(T) - \theta(0)| = 2\pi$. Therefore

$$2\pi = \Big| \int_0^T \dot{\theta}\, dt \Big| \leq \int_0^T |\dot{\theta}|\, dt \leq LT, \qquad \text{i.e.} \qquad T \geq \frac{2\pi}{L}. \qquad \square$$

Exercise 8.17

State an analogue of the Diamond Construction Lemma for flows in $\mathbb{R}^n$, $n \geq 3$.

Solution. Let $\mathbf{u}$ be a nonequilibrium point of $\dot{\mathbf{v}} = \mathbf{f}(\mathbf{v})$, $\mathbf{f} \in C^1$, $\mathbf{e} = \mathbf{f}(\mathbf{u})/|\mathbf{f}(\mathbf{u})|$, $a = |\mathbf{f}(\mathbf{u})|$. Let $H = \{\mathbf{u} + \mathbf{w} : \mathbf{w} \perp \mathbf{e}\}$ be the hyperplane through $\mathbf{u}$ perpendicular to $\mathbf{f}(\mathbf{u})$ (the $(n-1)$-dimensional transversal). Define the δ-diamond

$$D_\delta(\mathbf{u}) = \{\mathbf{u} + s\mathbf{e} + \mathbf{w} :\ \mathbf{w} \perp \mathbf{e},\ |s| + |\mathbf{w}| \leq \delta\}$$

(a double cone, with apexes at $\mathbf{u} \pm \delta\mathbf{e}$ and base the $(n-1)$-ball of radius δ in H; for $n = 2$ it is the square of Lemma 8.11).

Lemma. For $\delta > 0$ sufficiently small, if $\mathbf{v}_0$ is in the interior of $D_\delta(\mathbf{u})$, there are times $t_1 < 0 < t_2$ with $|t_i| \le 4\delta/a$ such that the trajectory segment $\{\mathbf{F}(t, \mathbf{v}_0) : t_1 \le t \le t_2\}$ lies in $D_\delta(\mathbf{u})$, begins on the back part of ∂D_δ ($s < 0$), ends on the front part ($s > 0$), and meets H in exactly one point.

Proof. Write $\mathbf{v}(t) = \mathbf{u} + s(t)\mathbf{e} + \mathbf{w}(t)$. Then $\dot{s} = \mathbf{e} \cdot \mathbf{f}(\mathbf{v})$ and $\dot{\mathbf{w}} = \mathbf{f}(\mathbf{v}) - \dot{s}\,\mathbf{e}$. Since $\mathbf{f}(\mathbf{u}) = a\mathbf{e}$, continuity gives δ_0 such that for $|\mathbf{v} - \mathbf{u}| \le \delta_0$: $\dot{s} > a/2$ and $|\dot{\mathbf{w}}| < a/2$; so $\dot{s} > |\dot{\mathbf{w}}|$. Take $\delta \le \delta_0$.

- s is strictly increasing while the orbit is in the diamond and ranges over an interval of length at most 2δ, so the orbit leaves in forward time within $2\delta/(a/2) = 4\delta/a$, and similarly in backward time.
- Let $h = |s| + |\mathbf{w}|$. Where $s > 0$: $\dot{h} \ge \dot{s} - |\dot{\mathbf{w}}| > 0$; where $s < 0$: $\dot{h} \le -\dot{s} + |\dot{\mathbf{w}}| < 0$. So in forward time h cannot reach δ while $s \le 0$ (it is decreasing there), hence the forward exit has $s > 0$. In backward time the reverse happens, so the backward exit has $s < 0$.
- Since s is strictly increasing, from $s(t_1) < 0$ to $s(t_2) > 0$, it vanishes exactly once, i.e. the segment meets H exactly once. □

Exercise 8.18

Let L be a transversal at a nonequilibrium point of a planar vector field. If an orbit crosses L at times t_1 and t_2, show it crosses L only finitely many times between t_1 and t_2.

Solution. Let $\mathbf{n}$ be a unit normal to the line containing L, and $h(t) = \mathbf{n} \cdot (\mathbf{v}(t) - \mathbf{u})$, where $\mathbf{u} \in L$. Then $\mathbf{v}(t) \in L$ implies $h(t) = 0$. Suppose there were infinitely many crossing times in $[t_1, t_2]$. By compactness they have an accumulation point $t^* \in [t_1, t_2]$, and since L is closed, $\mathbf{v}(t^*) =: \mathbf{p} \in L$ (it is a limit of points of L). Then $\mathbf{f}(\mathbf{p}) \neq 0$ (L contains no equilibria) and $\mathbf{f}(\mathbf{p})$ is not tangent to L, i.e. $\mathbf{n} \cdot \mathbf{f}(\mathbf{p}) \neq 0$. So $h(t^*) = 0$ and $h'(t^*) = \mathbf{n} \cdot \mathbf{f}(\mathbf{p}) \neq 0$, which means $h \neq 0$ on a punctured neighborhood of t^*. That contradicts infinitely many zeros of h (crossing times) accumulating at t^*. Hence there are finitely many crossings in $[t_1, t_2]$. □